QuickLoan Mobile Ethical Data Review

Your Role: Data Governance Consultant

Welcome to QuickLoan Mobile, a fast-growing fintech startup based in Ghana. They offer instant micro-loans via a mobile app, and their growth is driven by a fully automated Machine Learning (ML) loan-scoring model. Your role is to act as an **Independent Data Governance Consultant**. QuickLoan has hired you because an internal review flagged serious concerns about their data practices, specifically around **Governance, Quality, and Ethics**.

Your task is to review their existing data flow, identify the major risks, and propose practical, real-world solutions.

Project Objectives

By completing this project, you will demonstrate your ability to:

1. **Identify Risks:** Classify and analyze issues related to **Data Quality**, **Legal Compliance** (specifically Ghana's Data Protection Act, Act 843), and **Algorithmic Fairness/Bias**.
2. **Apply Principles:** Propose solutions using core principles like **Data Minimization**, **Consent Management**, **Data Classification**, and **Data Retention**.
3. **Drive Ethical Practice:** Recommend a practical, ethical reporting metric to ensure transparency in their automated decision-making process.

Scenario: The Flawed QuickLoan Data Pipeline

The company's data pipeline, shown in the **Flawed Data Flow Diagram** below, is the core of their governance problem. It contains **deliberate flaws** that you must identify.

Preliminary Concerns from Internal Audit:

- **Excessive Collection:** The app collects way more personal data than necessary for a loan decision (e.g., the user's entire contact list).
- **Compliance Risk:** There is no explicit consent capture and a high risk of violating Ghana's Data Protection Act (Act 843).

- **Data Quality:** Customer data is often incomplete or inconsistently formatted.
- **Algorithmic Bias:** The ML model's auto-approval decisions might be unfair to certain demographic groups.

Required Deliverables

You must submit the following three documents, which build on the analysis of the data flow:

Deliverable 1: Governance Review Card (Primary Task)

Complete the **governance_review_card** template below. You must provide three distinct risk findings and one reporting recommendation.

- **1. Data Quality Risk:** Identify one specific quality issue and its fix.
- **2. Legal & Compliance Risk:** Identify the PII classification and a major compliance risk related to Ghana's DPA.
- **3. Bias & Fairness Risk:** Pinpoint the source of potential bias and propose a monitoring step.
- **4. Storytelling/Reporting Metric:** Propose a single, actionable metric for ethical reporting.

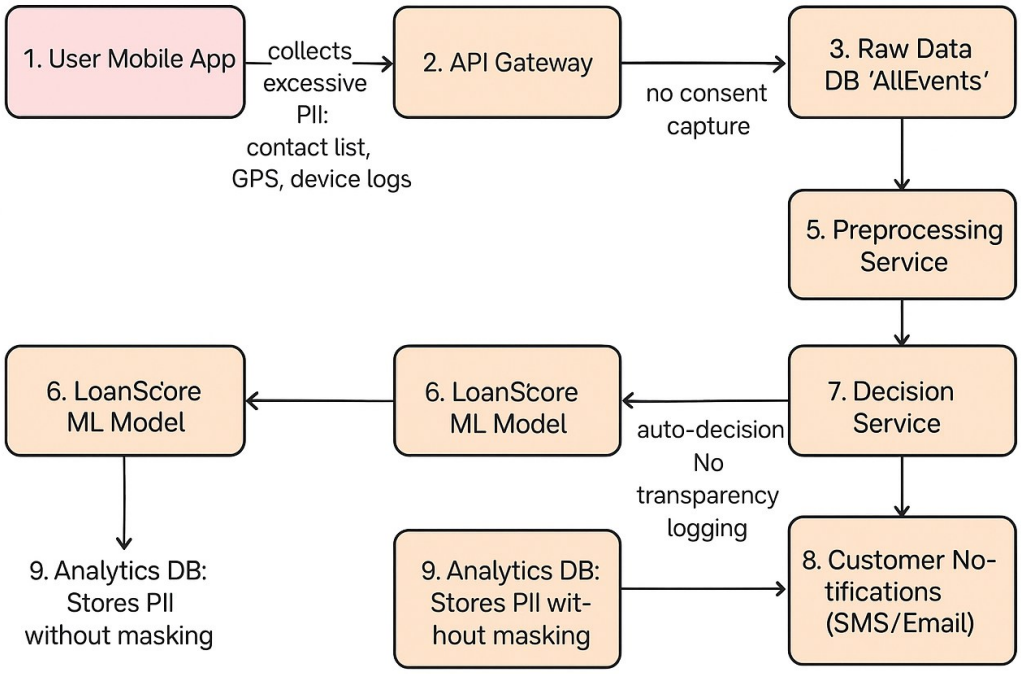
QuickLoan Governance Review Card

Complete the table below with concise, actionable findings and recommendations based on your review of the QuickLoan Mobile data flow diagram.

Section	**Issue/Definition**	**Impact**	**Suggested Fix/Mitigation**
1. Data Quality Risk			
2. Legal & Compliance Risk			
Data Classification (Choose one: Public / Internal / Confidential / Sensitive)			
3. Bias & Fairness Risk			
Source of Bias			
4. Storytelling / Reporting Recommendation			
Metric to Monitor (name & definition)			
Visualization Type (e.g., Line Chart, Grouped Bar Chart)			
Why It Matters (One sentence)			

Deliverable 2: Corrected Data Flow Diagram

Using the diagram provided in **data_flow_diagram** below, clearly indicate **at least three corrections**. For each correction, you must add an annotation (like a numbered footnote) that explains **what** change you made and **why** it is necessary (e.g., "Implement Data Masking: To protect PII in the Analytics DB from misuse.").



Key Flaws to Identify and Correct:

1. **Excessive Collection** at Step 1 (User Mobile App).
2. **No Consent/Compliance** between Step 2 and Step 3 (API Gateway to Raw Data DB).
3. **No Classification/Retention** at Step 3 (Raw Data DB).
4. **No Defined Handling** at Step 4 (Preprocessing Service).
5. **No Transparency/Logging** at Step 7 (Decision Service).
6. **No Masking/Anonymization** at Step 9 and 10 (Analytics DB and 3rd-Party Partner).

Deliverable 3: Summary of Review Process (200-300 Words)

Write a short essay (200-300 words) summarizing your process. Address the following:

- How did you use **Data Lifecycle** and **Classification** principles to find the risks?
- How does your proposed **Metric** (from Deliverable 1) ensure ethical and transparent governance?